

Infinitesimal Calculus 2 — Lecture 4 (18.03.2026)

Differentiability and Partial Derivatives

Partial Derivatives — Formal Definitions

For $z = f(x, y)$ at (x_0, y_0) , the increments are:

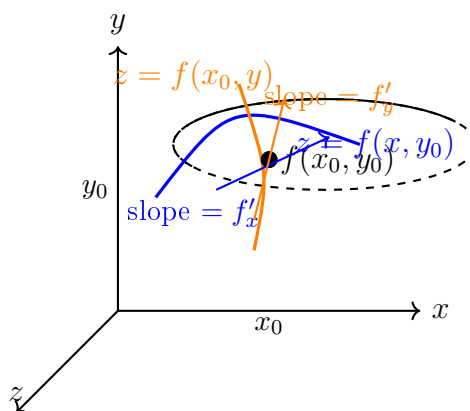
$$\Delta_x f(x_0, y_0) = f(x_0 + \Delta x, y_0) - f(x_0, y_0), \quad \Delta_y f(x_0, y_0) = f(x_0, y_0 + \Delta y) - f(x_0, y_0).$$

Definition 1 (Partial derivatives).

$$f'_x(x_0, y_0) = \lim_{\Delta x \rightarrow 0} \frac{\Delta_x f(x_0, y_0)}{\Delta x} = \lim_{\Delta x \rightarrow 0} \frac{f(x_0 + \Delta x, y_0) - f(x_0, y_0)}{\Delta x},$$

$$f'_y(x_0, y_0) = \lim_{\Delta y \rightarrow 0} \frac{\Delta_y f(x_0, y_0)}{\Delta y} = \lim_{\Delta y \rightarrow 0} \frac{f(x_0, y_0 + \Delta y) - f(x_0, y_0)}{\Delta y}.$$

Geometric meaning of partial derivatives



$f'_x(x_0, y_0)$ is the slope of the cross-section $z = f(x, y_0)$ (blue curve) at x_0 .
 $f'_y(x_0, y_0)$ is the slope of the cross-section $z = f(x_0, y)$ (orange curve) at y_0 .

Differentiability

Definition 2 (Differentiable function). $z = f(x, y)$ is **differentiable** at (x_0, y_0) if:

$$\Delta f(x_0, y_0) = A \Delta x + B \Delta y + \alpha \Delta x + \beta \Delta y, \quad \alpha, \beta \rightarrow 0 \text{ as } \Delta x, \Delta y \rightarrow 0.$$

The **total differential** is $df(x_0, y_0) = A \Delta x + B \Delta y$.

Setting $\Delta \rho = \sqrt{(\Delta x)^2 + (\Delta y)^2}$ and writing $\varepsilon \cdot \Delta \rho = \alpha \Delta x + \beta \Delta y$:

$$\varepsilon = \underbrace{\alpha \cdot \frac{\Delta x}{\Delta \rho}}_{\cos \theta} + \underbrace{\beta \cdot \frac{\Delta y}{\Delta \rho}}_{\sin \theta} \rightarrow 0 \quad \text{as } \Delta \rho \rightarrow 0.$$

Main Theorems

Theorem 1 (Thm 1: Differentiable $\Rightarrow$ Continuous). *If f is differentiable at (x_0, y_0) , then f is continuous there.*

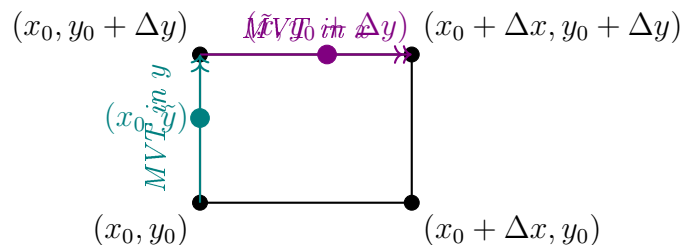
Proof: $\Delta f = A \Delta x + B \Delta y + \alpha \Delta x + \beta \Delta y \rightarrow 0$.

Theorem 2 (Thm 2: Differentiable $\Rightarrow$ Partial derivatives exist; $A = f'_x$, $B = f'_y$). *Set $\Delta y = 0$: $\Delta_x f = (A + \alpha) \Delta x$. Dividing by Δx and taking the limit gives $f'_x = A$. Similarly $f'_y = B$.*

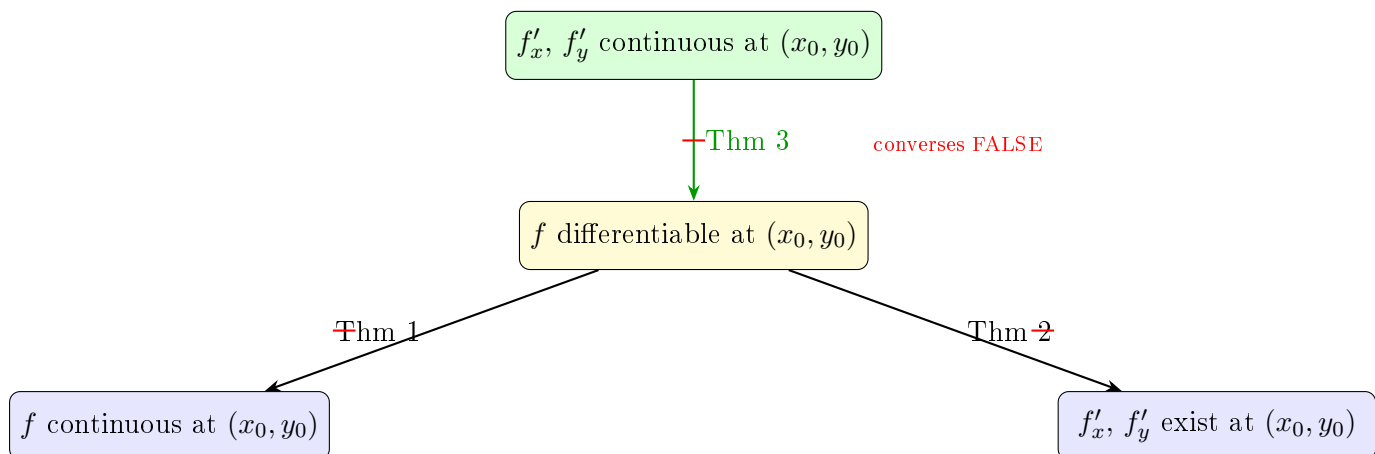
Theorem 3 (Thm 3: Continuous partials $\Rightarrow$ Differentiable). *If f'_x, f'_y exist and are continuous at (x_0, y_0) , then f is differentiable there.*

Proof sketch (Lagrange MVT):

$$\begin{aligned} \Delta f &= \underbrace{f(x_0 + \Delta x, y_0 + \Delta y) - f(x_0, y_0 + \Delta y)}_{\text{MVT on } x} + \underbrace{f(x_0, y_0 + \Delta y) - f(x_0, y_0)}_{\text{MVT on } y} \\ &= f'_x(\tilde{x}, y_0 + \Delta y) \Delta x + f'_y(x_0, \tilde{y}) \Delta y \\ &= (f'_x(x_0, y_0) + \alpha) \Delta x + (f'_y(x_0, y_0) + \beta) \Delta y, \quad \alpha, \beta \rightarrow 0. \end{aligned}$$



Hierarchy of Properties



Examples

Example 1 (Differentiable — check from definition). $f(x, y) = x^2 - xy$ at $(1, 4)$.

$$\begin{aligned} \Delta f(1, 4) &= (1 + \Delta x)^2 - (1 + \Delta x)(4 + \Delta y) - [1 - 4] \\ &= -2\Delta x - \Delta y + \underbrace{(\Delta x)^2}_{\alpha = \Delta x} - \underbrace{\Delta x \Delta y}_{\beta = -\Delta y}. \end{aligned}$$

So f is differentiable at $(1, 4)$ and $df(1, 4) = -2\Delta x - \Delta y$.

Check: $f'_x = 2x - y \Rightarrow f'_x(1, 4) = -2$; $f'_y = -x \Rightarrow f'_y(1, 4) = -1$. ✓

Example 2 (Partial derivatives exist, NOT continuous, NOT differentiable).

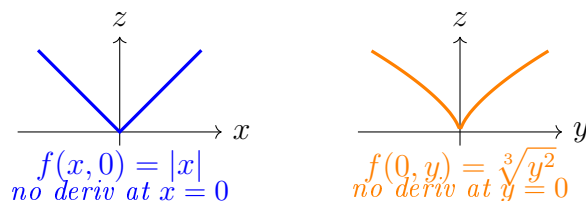
$$f(x, y) = \begin{cases} \frac{xy}{x^2 + y^2}, & (x, y) \neq (0, 0), \\ 0, & (x, y) = (0, 0). \end{cases}$$

$f'_x(0, 0) = \lim_{\Delta x \rightarrow 0} \frac{\Delta x \cdot 0 / (\Delta x)^2 - 0}{\Delta x} = 0$, similarly $f'_y(0, 0) = 0$.

But along $x = y \rightarrow 0$: $f \rightarrow \frac{1}{2} \neq 0$. So f is **not continuous**, hence **not differentiable** at $(0, 0)$.

Example 3 (Continuous but NO partial derivatives).

$$f(x, y) = |x| + \sqrt[3]{y^2}.$$



f is continuous at $(0, 0)$ but $f'_x(0, 0)$ and $f'_y(0, 0)$ do **not exist**.

Example 4 (Differentiable, but partials NOT continuous).

$$f(x, y) = \begin{cases} (x^2 + y^2) \sin \frac{1}{x^2 + y^2}, & (x, y) \neq (0, 0), \\ 0, & (x, y) = (0, 0). \end{cases}$$

Continuous at $(0, 0)$: $|f| \leq \rho^2 \rightarrow 0$.

Partial derivatives at $(0, 0)$: $f'_x(0, 0) = \lim_{\Delta x \rightarrow 0} \Delta x \sin \frac{1}{(\Delta x)^2} = 0$. Similarly $f'_y(0, 0) = 0$.

Differentiability check:

$$\varepsilon = \frac{\Delta f - 0 \cdot \Delta x - 0 \cdot \Delta y}{\Delta \rho} = \frac{(\Delta \rho)^2 \sin \frac{1}{(\Delta \rho)^2}}{\Delta \rho} = \Delta \rho \cdot \underbrace{\sin \frac{1}{(\Delta \rho)^2}}_{\text{bounded}} \rightarrow 0.$$

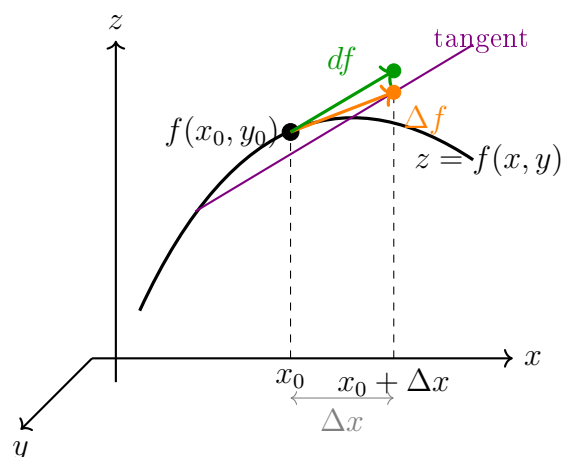
So f is differentiable at $(0, 0)$.

Partials not continuous: For $(x, y) \neq (0, 0)$:

$$f'_x(x, y) = 2x \sin \frac{1}{x^2 + y^2} - \frac{2x}{x^2 + y^2} \cos \frac{1}{x^2 + y^2}.$$

Along $y = 0$, $x \rightarrow 0$: second term $\sim \frac{2}{x} \cos \frac{1}{x^2}$ is unbounded. Hence f'_x is **not continuous** at $(0, 0)$.

Geometric Interpretation of the Differential



$$\Delta f(x_0, y_0) = f(x_0 + \Delta x, y_0 + \Delta y) - f(x_0, y_0) \quad (\text{actual increment, orange})$$

$$df(x_0, y_0) = f'_x(x_0, y_0) \Delta x + f'_y(x_0, y_0) \Delta y \quad (\text{linear approximation along tangent plane, green})$$

Tangent Plane

The **tangent plane** to $z = f(x, y)$ at (x_0, y_0, z_0) where $z_0 = f(x_0, y_0)$:

$$z - z_0 = f'_x(x_0, y_0) (x - x_0) + f'_y(x_0, y_0) (y - y_0)$$